

Estimating Unpaid Losses for a Workers' Compensation Book

Chain-Ladder and Bornhuetter-Ferguson reserve estimates with Mack uncertainty

Nigel Mpuah | Data: CAS Loss Reserve Database (NAIC Schedule P) | Federal Insurance Company (Chubb)

Overview

A property-casualty insurer collects premium up front but pays claims out over years — in workers' compensation, sometimes over a decade as medical and wage-replacement payments run their course. At any point the company has paid out part of what it owes and still carries the rest as a liability on claims that have already happened. Estimating that unpaid amount, the *reserve*, is the core work of a reserving actuary. Set it too low and the company looks profitable until the shortfall surfaces; set it too high and capital sits idle that could have been deployed.

This analysis estimates unpaid losses for one workers' compensation book using two standard methods and a measure of the uncertainty around the estimate. I work from real regulatory data, build a paid-loss development triangle, and project each accident year to its ultimate cost. The headline result is a chain-ladder reserve of **\$381.6M**, with a Mack coefficient of variation near **16%**. The Bornhuetter-Ferguson comparison lands much higher under a naive assumption, and the gap — along with its fix — turns out to be the most instructive part of the exercise.

Data

The data comes from the CAS Loss Reserve Database, which the Casualty Actuarial Society compiled from insurers' NAIC Schedule P regulatory filings. I use the workers' compensation file and isolate a single carrier, **Federal Insurance Company** (a Chubb subsidiary, group code 388). Picking one company keeps the development pattern coherent rather than blending unrelated books. Federal is a good choice for this: it writes a large, stable book, and its paid losses develop smoothly and stay positive throughout — which is exactly what the methods below need.

Each record gives cumulative paid losses for an accident year at a given development lag, along with earned premium. Accident years span 1998 through 2006 in the realized data, each followed for up to ten development periods. Arranging paid losses by accident year (rows) and development lag (columns) produces the development triangle that everything else is built on.

Method

The development triangle

The triangle records how each accident year's paid losses grow as the years mature. Reading across a row shows one accident year paying out over time; reading down a column compares the same stage of maturity across years. The upper-left is fully developed — old years that have had a decade to settle — while the lower-right is blank, because recent years haven't had time to play out yet. That empty corner is what reserving estimates.

Chain-ladder

Chain-ladder assumes each accident year develops like the others did. For every pair of adjacent development columns I compute an age-to-age factor — the volume-weighted ratio of losses at the later age to losses at the earlier age — which captures the typical growth from one period to the next. Multiplying the remaining factors together gives a cumulative development factor (CDF) that grosses up an accident year’s paid-to-date figure to its estimated ultimate. The reserve is ultimate minus paid-to-date. No tail factor is applied, since the oldest year is effectively settled by age ten.

Mack standard error

A single reserve number says nothing about how reliable it is. Mack’s method puts a standard error around the chain-ladder estimate using only the triangle itself, with no distributional assumptions beyond the chain-ladder model. It lets me express each accident year’s reserve as a central estimate plus a band, and summarize overall reliability with a coefficient of variation. I also use it to cross-check that my hand-computed ultimates agree with an independent implementation.

Bornhuetter-Ferguson

Bornhuetter-Ferguson blends the data-driven projection with an external expectation. Instead of grossing up the paid figure alone, it applies an *a priori* expected ultimate — premium times an expected loss ratio (ELR) — to only the unreported share of losses, the part chain-ladder is least sure about. Mature years, which are nearly fully reported, end up close to their paid figure; green years lean on the *a priori* rather than extrapolating from a thin early number. The method is only as good as that ELR, a point the results make vivid.

Results

Chain-ladder reserves

The table below shows the chain-ladder projection by accident year. The CDF climbs as years get greener: 1998 is fully mature at a factor of 1.00, while 2006, seen at only its second development period, needs a $2.09\times$ gross-up. The reserve concentrates in recent years for the same reason — they have the most development still ahead.

Accident Yr	Paid	Age	CDF	Ultimate	IBNR
1998	242,368	10	1.000	242,368	0
1999	251,726	9	1.017	255,993	4,267
2000	238,568	8	1.041	248,225	9,657
2001	178,283	7	1.078	192,258	13,975
2002	191,706	6	1.101	211,018	19,312
2003	194,767	5	1.176	228,951	34,184
2004	198,619	4	1.309	260,001	61,382
2005	175,099	3	1.543	270,238	95,139
2006	132,336	2	2.086	276,019	143,683
Total					381,599

Table 1: Volume-weighted chain-ladder projection. Figures in \$000s. Total reserve \$381.6M.

Plotting the per-year reserve with Mack standard-error bands makes the pattern and its uncertainty visible together. The reserve grows toward recent accident years, and so does the uncertainty — the error bars widen because greener years are projected from less data through larger factors. The overall coefficient of variation is roughly 16%, a reasonable level for a workers' compensation book, and the Mack ultimates matched my hand-computed figures to within a small uniform gap attributable to the trimmed tail period.

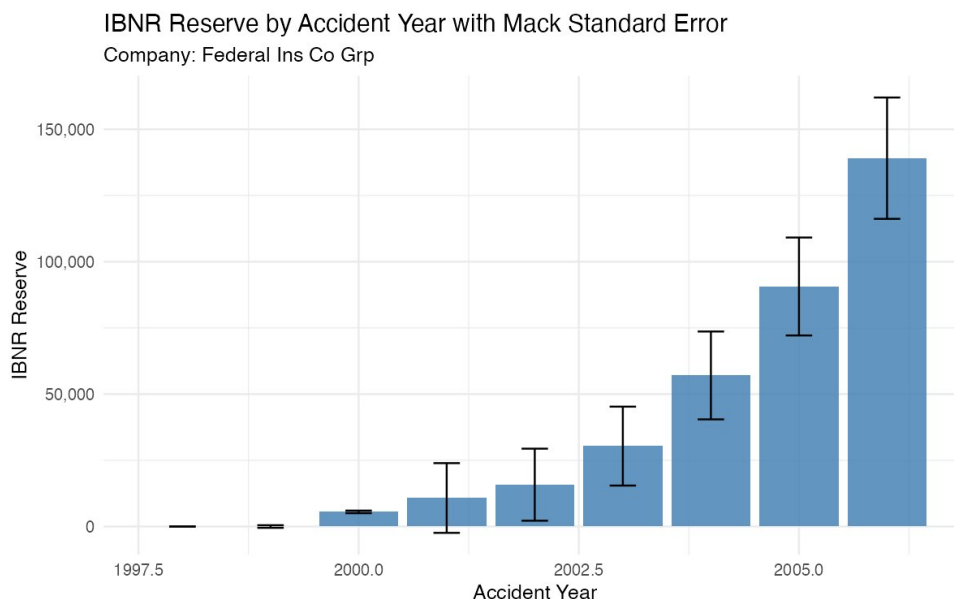


Figure 1: Chain-ladder IBNR by accident year with $\pm$ one Mack standard error.

Chain-ladder versus Bornhuetter-Ferguson

I first derive the ELR from the mature accident years — those essentially fully developed, where paid losses are a clean read on ultimate — which gives 63.7%. Applying Bornhuetter-Ferguson with that assumption produces the comparison below.

Accident Yr	Net premium	% reported	CL IBNR	BF IBNR
1998	367,007	100.0%	0	0
1999	379,679	98.3%	4,267	4,031
2000	403,622	96.1%	9,657	10,002
2001	514,834	92.7%	13,975	23,835
2002	512,942	90.8%	19,312	29,899
2003	648,372	85.1%	34,184	61,659
2004	842,351	76.4%	61,382	126,662
2005	925,952	64.8%	95,139	207,630
2006	864,195	47.9%	143,683	286,529
Total			381,599	750,247

Table 2: CL and naive-BF reserves by accident year. Net premium and IBNR in \$000s; % reported from the chain-ladder CDF.

Discussion

The two methods agree closely on the mature years and diverge sharply on the green ones. For 1999 and 2000 the reserves are nearly identical; by 2006 chain-ladder estimates \$144M while Bornhuetter-Ferguson estimates \$287M. In total, naive BF comes in at \$750M against chain-ladder's \$382M — nearly double. That is the opposite of BF's usual role as the more stable, conservative method, and the reason is worth spelling out.

The cause is the a priori. The ELR of 63.7% is derived from 1998–99, the only accident years mature enough to qualify, and those early years carried genuinely high loss ratios. Meanwhile this book's premium more than doubled over the period, from \$367M in 1998 to over \$900M in 2005. Bornhuetter-Ferguson applies that single high, stale loss ratio to all the grown premium, and to the large unreported share of recent years — inflating their reserves well past what the actual development supports. The paid data is signaling that recent accident years are running toward a lower loss ratio than the late-1990s years did, and the naive BF a priori simply overrides that signal.

Correcting the a priori

The fix is to anchor the ELR to something representative of the whole book rather than two old years. Using the chain-ladder ultimates themselves — total projected ultimate over total premium — gives a CL-implied ELR of **40.0%**, far below the naive 63.7%. That figure reflects how this book has actually performed across all nine accident years, not just its earliest two.

Re-running Bornhuetter-Ferguson with the 40.0% a priori brings the total reserve down to **\$471M**, far closer to chain-ladder's \$382M. More importantly, the per-year behavior is now correct: BF sits just above chain-ladder on the green years (2006: \$180M versus \$144M) rather than doubling it. That is exactly what the method is designed to do — lean modestly on the a priori where chain-ladder is grossing up thin early data, without overriding the experience entirely.

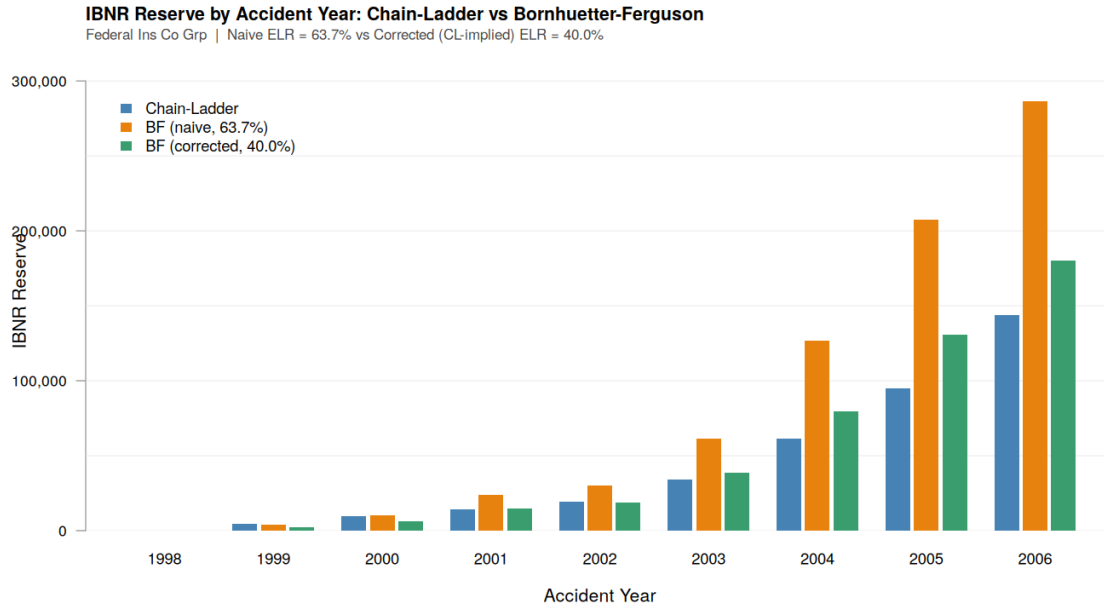


Figure 2: IBNR by accident year: chain-ladder, naive BF (63.7%), and corrected BF (40.0%). The corrected a priori pulls BF back alongside chain-ladder.

This is the central lesson of the comparison. Bornhuetter-Ferguson is exactly as good as its expected loss ratio; anchoring that ratio to a small, unrepresentative slice of history biases every estimate built on it, while a representative a priori restores the method’s intended stabilizing behavior. Taken together, the chain-ladder reserve of \$381.6M is the credible central estimate for this book, supported by a Mack coefficient of variation near 16% and corroborated by a properly calibrated Bornhuetter-Ferguson at \$471M.

Notes on implementation

The analysis is implemented in R. Data handling and plotting use the tidyverse; the Mack cross-check uses the ChainLadder package, while the triangle, development factors, and reserves are computed directly so the mechanics are explicit rather than hidden behind a function call. Because Mack’s method requires a square triangle, the final development column — holding only the fully mature 1998 cell — is trimmed before that step, which has no material effect on the estimates. All figures are in thousands of dollars, consistent with the source data.